

NEW WIRELESS TECHNOLOGIES To Help Meet Aviation Demands

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This article discusses some key recent developments in the area of wireless networking in the aviation industry



(Credit: Google Images)

New techniques and technologies in wireless communication are needed for safe and efficient aviation operations. To keep track of airplanes, pilots use high-frequency radio signals to communicate with ground controllers and other aircraft. The aircraft in low-altitude airspace plays a key role in this network, for prospects such as emergency communication, routing patrol and wireless data relay.

A unique application of wireless technology came into focus when Airbus H145 manned helicopter and Schiebel Camcopter S-100 unmanned aerial vehicle (UAV) demonstrated the highest level of Manned Unmanned Teaming (MUM-T) capabilities. The manned helicopter had full control over the UAV, including its take off, landing, payload, launch, recovery and transmission of sensor data obtained by the latter.

A smaller unmanned aerial system (UAS) having vertical take-off and landing capabilities can fly closer around obstacles, such as trees or buildings, to detect hidden objects in places not accessible by traditional helicopters. It can explore unknown

territory and deliver information to the helicopter crew, which is operating from a safe position and which can then step in with the helicopter's superior effects, having received a clear picture from the UAS.

This wireless technology advancement is highly valuable and mission enhancing for army aviation in complex operations and dangerous environments. MUM-T leverages the strengths of both manned and unmanned systems by providing pilots of manned aircraft with the ability to take full advantage of the intelligence, surveillance and reconnaissance (ISR) capabilities of the UAS and, thereby, significantly improving safety and decision-making capability in complex missions. This approach of enhancing coverage and timeliness of information, while keeping pilots and manned assets safe, enables commanders to maximise the advantages offered by both the platforms.

Role of wireless technology in aviation

Wireless technologies and networks are playing an important role in aviation at a